Day 3: BIOS and POST Troubleshooting Theory (30 min): Understanding BIOS and POST errors. Practical (30 min): Analyze POST error codes on a given setup.

1.List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.

ANS:

No.	BIOS/POST Error Message	Possible Hardware or Configuration Issue
1	CMOS Battery Failure	The motherboard CMOS battery may be weak or dead.
2	Keyboard Not Detected	The keyboard may be disconnected, faulty, or connected to a faulty USB port.
3	No Bootable Device Found	The hard drive or SSD may not be detected, or the boot order may be incorrect.
4	Memory Error	The RAM may be loose, faulty, or incompatible with the motherboard.
5	CPU Fan Error	The CPU fan may be disconnected, faulty, or running too slowly.

2.Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.

ANS:

Meaning: In many AMI BIOS systems, 3 beeps usually mean there is a RAM (memory) problem.

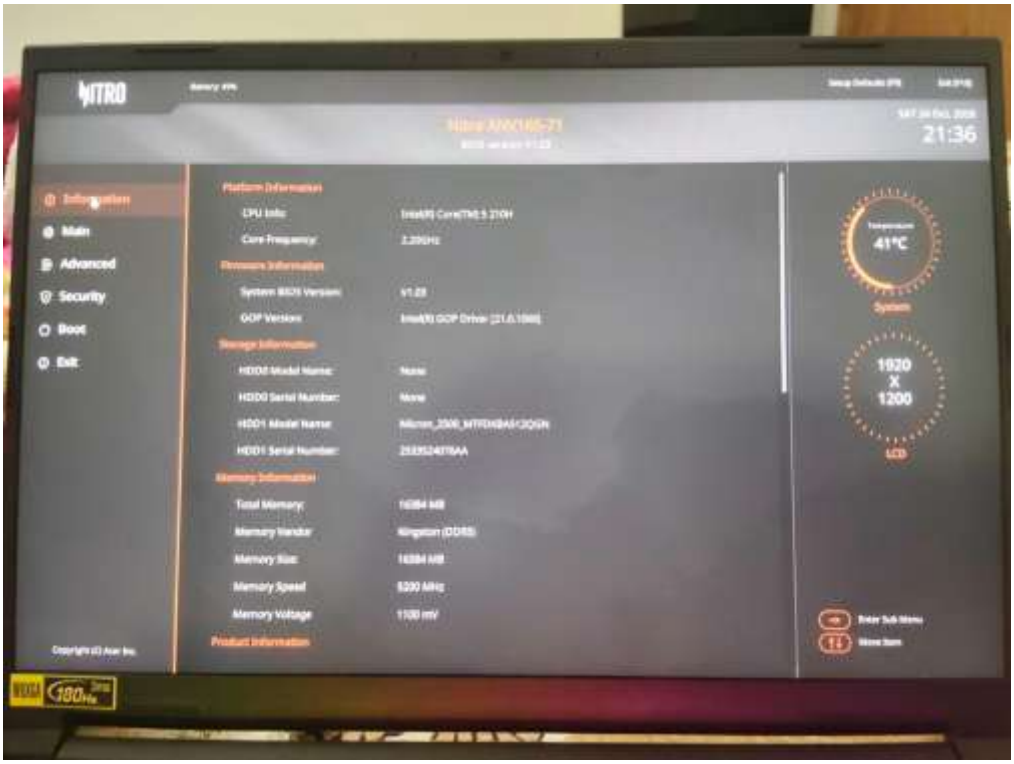
Two Troubleshooting Steps

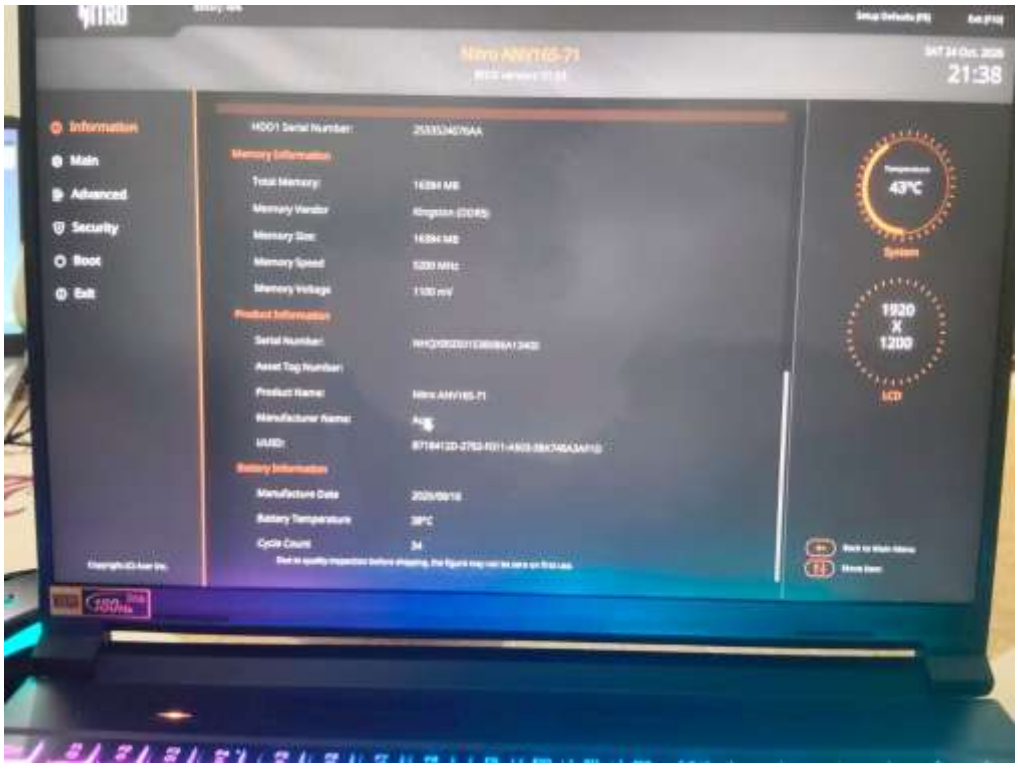
1. Check the RAM: Turn off the PC, unplug it, remove the RAM, and put it back properly.

2. Test the RAM: Try one RAM stick at a time or use another working, compatible RAM stick.

3.Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.

ANS: The motherboard manufacturer of the device shown in the photos below is acer and the BIOS version is V1.03





**4.You are given a screenshot of a POST error message that says 'CMOS checksum error - Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system.

Hint: Think about the CMOS battery and BIOS settings reset.**

ANS:



Restart the PC: Turn the computer off and on again.

Open BIOS: Press F2 or Delete when the computer starts.

Set date and time: Correct the date and time in BIOS.

Load default settings: Select “Load Setup Defaults” if needed.

Save settings: Press F10 to save and exit, if that is the correct key for your BIOS.

Check the CMOS battery: If the error appears again, the battery may be weak.

Replace the battery: Turn off and unplug the PC, then replace the CMOS battery with the correct type, often a CR2032.

Restart the PC: Set the date and time again and check whether the error is gone.